PROGRAM-5

1. Input Data

User Requirements:

- Monthly Data Usage (GB)
- Call Minutes
- Required Validity (Days)
- Budget Limit

Mobile Plan Details:

- Plan Name
- Price
- Data Limit
- Call Limit
- Validity
- Excess Data Charge
- Excess Call Charge

2. Eligibility Check

A plan is considered **eligible** if:

- Validity ≥ Required Validity
- Price ≤ Budget
- Supports required usage levels

If a plan does not fully cover usage, excess charges are calculated.

3. Excess Charge Calculation

Extra Data Cost

$$\text{Extra Data Cost} = (\text{Required Data} - \text{Plan Data}) \times \text{Data Charge Rate}$$

Extra Call Cost

$$\text{Extra Call Cost} = (\text{Required Minutes} - \text{Plan Minutes}) \times \text{Call Charge Rate}$$

Total Cost

$$\text{Total Cost} = \text{Plan Price} + \text{Extra Data Cost} + \text{Extra Call Cost}$$

4. Plan Selection

Compare all eligible plans and select the plan with the **lowest total cost**.

5. Algorithm

1. Input user requirements.
2. Read available mobile plans.
3. Check validity and budget constraints.
4. Calculate excess data and call charges for each plan.
5. Determine total cost for every eligible plan.
6. Compare total costs.
7. Select the most economical plan.
8. Display plan comparison and recommendation.

6. Example

User Requirements

- Data Needed = 120 GB
- Call Minutes = 1000
- Validity = 28 Days
- Budget = ₹450

Available Plans

Plan	Price	Data	Calls	Validity
A	₹299	80 GB	Unlimited	28 Days
B	₹399	120 GB	Unlimited	28 Days
C	₹449	150 GB	Unlimited	56 Days

Assume:

- Extra Data Charge = ₹5 per GB

Cost Analysis

Plan A

- Extra Data = 40 GB
- Extra Cost = ₹200
- Total Cost = ₹499

Plan B

- Extra Data = 0
- Total Cost = ₹399

Plan C

- Extra Data = 0
- Total Cost = ₹449

Output

Eligible Plans:
Plan B : ₹399
Plan C : ₹449

Most Economical Plan:
Plan B

Total Cost: ₹399
Savings compared to next best plan: ₹50

programm

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# Multi-Plan Telecommunications Cost Optimization

# Input user requirements
data_required = float(input("Enter required data (GB): "))
calls_required = int(input("Enter required call minutes: "))
validity_required = int(input("Enter required validity (days): "))
budget = float(input("Enter your budget (Rs.): "))
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# Mobile plans: name, price, data, calls, validity, excess data rate, excess call rate
plans = [
    ("Plan A", 199, 2, 300, 28, 50, 0.50),
    ("Plan B", 299, 5, 600, 28, 40, 0.40),
    ("Plan C", 399, 10, 1000, 56, 30, 0.30),
    ("Plan D", 499, 15, 1500, 84, 25, 0.25)
]

eligible_plans = []

print("\n---------- MOBILE PLAN COMPARISON ----------")

for plan in plans:
    name, price, data, calls, validity, data_rate, call_rate = plan

    # Calculate excess usage charges
    excess_data = max(0, data_required - data)
    excess_calls = max(0, calls_required - calls)

    excess_charge = (excess_data * data_rate) + (excess_calls * call_rate)
    total_cost = price + excess_charge

    # Check validity, budget, and usage constraints
    if validity >= validity_required and total_cost <= budget:
        if data >= data_required and calls >= calls_required:
            eligible_plans.append((name, total_cost, price))

    print(f"\nPlan: {name}")
    print(f"Base Price: Rs. {price:.2f}")
    print(f"Included Data: {data} GB")
    print(f"Included Calls: {calls} minutes")
    print(f"Validity: {validity} days")
    print(f"Excess Charges: Rs. {excess_charge:.2f}")
    print(f"Total Cost: Rs. {total_cost:.2f}")

# Find the most economical eligible plan
print("\n---------- RESULT ----------")

if eligible_plans:
    best_plan = min(eligible_plans, key=lambda x: x[1])

    print("Most Economical Eligible Plan:", best_plan[0])
    print("Plan Price: Rs.", format(best_plan[2], ".2f"))
    print("Total Cost: Rs.", format(best_plan[1], ".2f"))
    print("Remaining Budget: Rs.", format(budget - best_plan[1], ".2f"))
```

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else:
    print("No mobile plan satisfies all your requirements and budget.")
```

output:

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Enter required data (GB): 986
Enter required call minutes: 546
Enter required validity (days): 235
Enter your budget (Rs.): 78

Plan: Plan A
Base Price: Rs. 199.00
Excess Charges: Rs. 49323.00
Total Cost: Rs. 49522.00

Plan: Plan B
Base Price: Rs. 299.00
Excess Charges: Rs. 39240.00
Total Cost: Rs. 39539.00

Plan: Plan C
Base Price: Rs. 399.00
Excess Charges: Rs. 29280.00
Total Cost: Rs. 29679.00

Plan: Plan D
Base Price: Rs. 499.00
Excess Charges: Rs. 24275.00
Total Cost: Rs. 24774.00

========== FINAL RESULT ==========
No eligible plan satisfies all requirements.

Process finished with exit code 0
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